Department of Engineering Science
School of Science & Technology



CES 440 Data Communications
Ch.2 The Physical Layer Part II

Dr. Jingxian Wu
(707) 664-3491
Jingxian.wu@sonoma.edu

OUTLINE

PART I

- **Functions of Physical Layer**
- **Theoretical Basis for Physical Layer**
- **Guided Transmission Media**
- **Terrestrial Wireless Transmission**
- **Communication Satellite**

PART II

- **Public Switched Telephone Network (PSTN)**
- **Mobile Telephone System**
- **Cable Television**

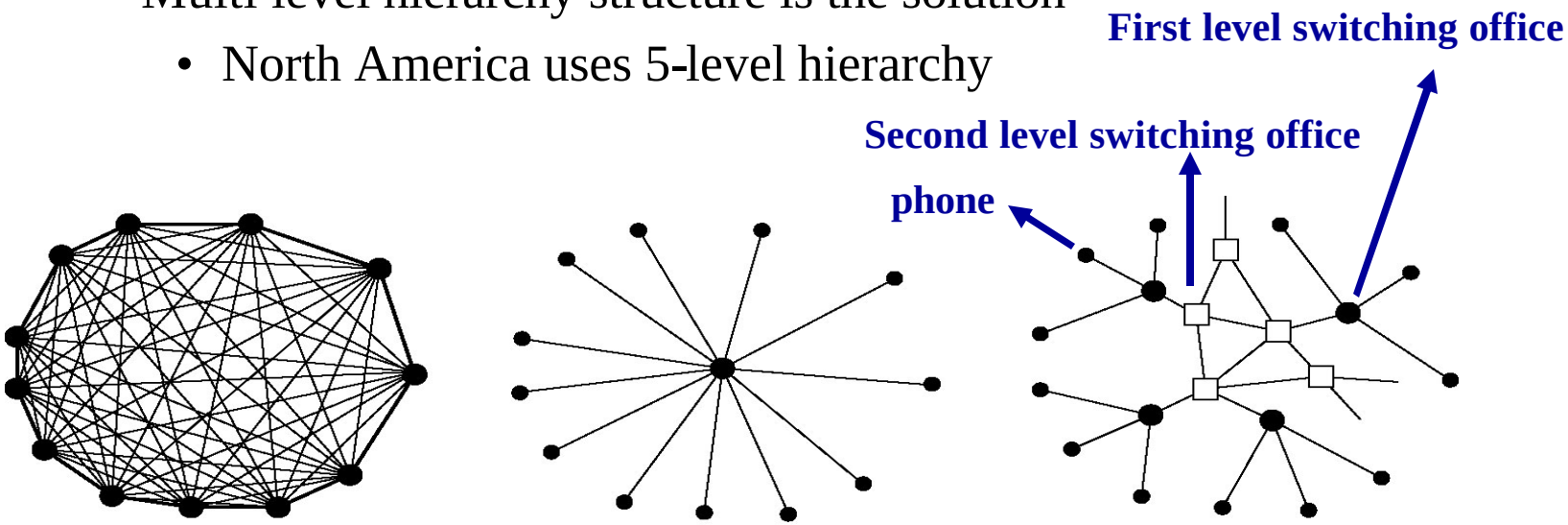
PUBLIC SWITCHED TELEPHONE NETWORK

- **Long distance communication between two computers usually rely on existing telecommunication facilities.**
  - Short distance computer communication can be implemented by running private cable between two computers.
  - It's economic prohibitive to run cable through two computers far away.

- **Public switched telephone network (PSTN)**
  - Dial-up, DSL
  - Backbone
    - most long distance transmission of telephone system is using fiber optics

PSTN: STRUCTURE

- **Multi-level hierarchy structure**
  - Fully-interconnected network is not scalable
  - Centralized switch model is better
    - But still not scalable for large amount of customers
  - Multi-level hierarchy structure is the solution
    - North America uses 5-level hierarchy



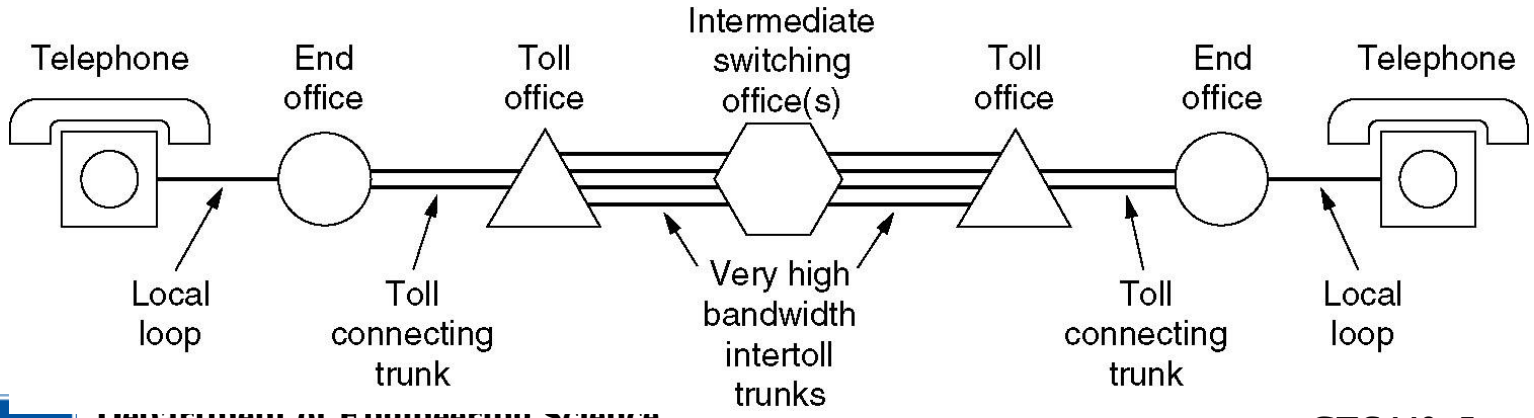
(a) Fully-interconnected network (1876)

(b) Centralized switch (1878)

(c) Two-level hierarchy (1890)

PSTN: STRUCTURE

- **Switching Offices**
  - End office: connected directly to subscriber (1st level)
  - Toll office (Tandem office): each end office is connected to several toll offices (2nd level)
  - Intermediate switching offices: (3rd level or higher)
- **Local loop (weakest link)**
  - Analog twisted pair connecting telephone and end office.
- **Trunks**
  - Digital fiber optics connecting switching offices

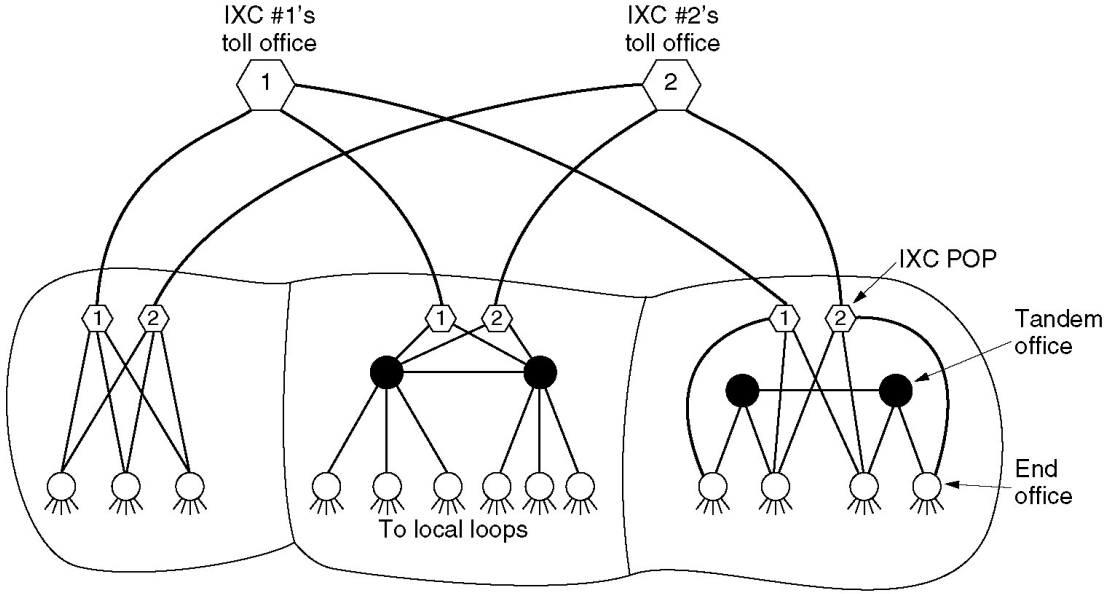


PSTN: OPERATIONS IN U.S.

- **U.S. is divided into 164 LATA**
  - LATA: Local Access and Transport Areas
    - The area covered by one area code
  - In each LATA, there is an LEC (Local Exchange Carrier)
    - Service provider for traditional telephone (In our area, AT&T)
  - Inter-LATA traffic (long distance) is handled by IXC
    - InterXchange Carrier, e.g. AT&T, Sprint, MCI, etc.

• POP: Point of Presence

• With POP, Subscriber can choose IXC freely

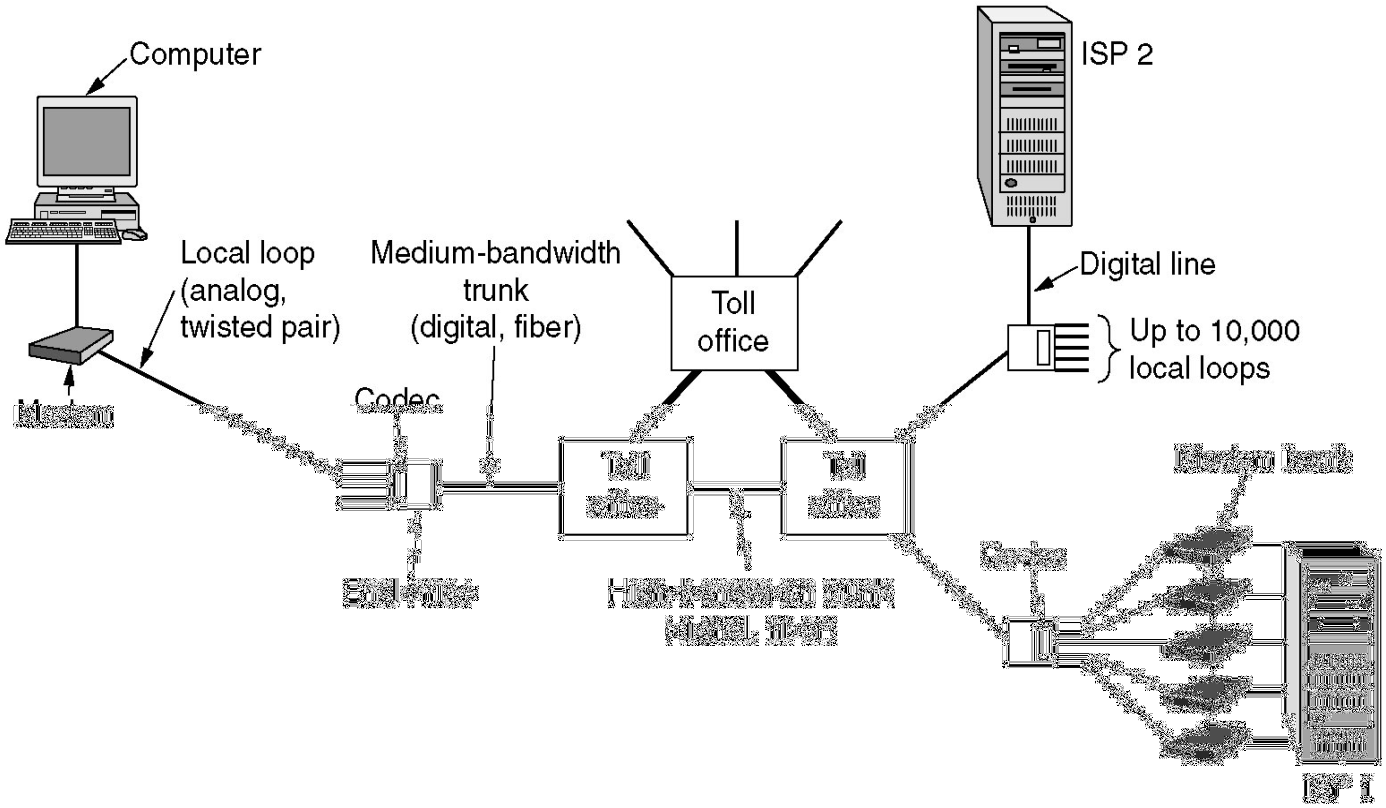


PSTN: LOCAL LOOP

- **Local loop**
  - Twisted pair connecting subscriber phones and end office
  - Often referred to as “last mile” or “access network”
- **An end office usually has 10,000 local loops**
  - Area code + exchange (707-664-xxxx) was used to indicate an end office
  - Recently, more complicated mapping schemes are introduced.
- **Data communication in local loop**
  - Dial-up modem
  - DSL (Digital subscriber lines)
  - ISDN (Integrated service digital network)

PSTN: DIAL-UP MODEM

- **Modem**
  - Modulator and Demodulator
- **Example: A computer using dial-up to access internet**



PSTN: DIAL-UP MODEM

- **BW available to each user: 4,000Hz**
- **V.32 modem**
  - Modulation: 32-QAM ➔ 5 bits/symbol (4 bits data, 1 bit parity check)
  - Baud: 2400 symbols/second
  - Data rate: 4 x 2400 = 9600 bps
- **V.34 bis modem**
  - Modulation: 14 data bits/symbol
  - Baud: 2400 symbols/second
  - Data rate: 14 x 2400 = 33,600 bps
  - When both user and ISP use analog lines, the upper limit is 35kbps.
- **V.92 modem**
  - When ISP end use digital lines, the upper limit is about 70kbps.
  - PCM Modulation: 8000 samples/second, 8 bits/symbol (7 bits data, 1 bit parity check)
  - Data rate: 8000 x 7 = 56,000 bps.

PSDN: DIGITAL SUBSCRIBER LINES

- **Can we achieve a high data rate with the old plain telephone line?**
  - At end office, there is a filter at the end of local loop
    - Attenuates all frequencies below 300Hz and above 3400Hz (3dB points)
    - The null-null BW is approximately 4000Hz.

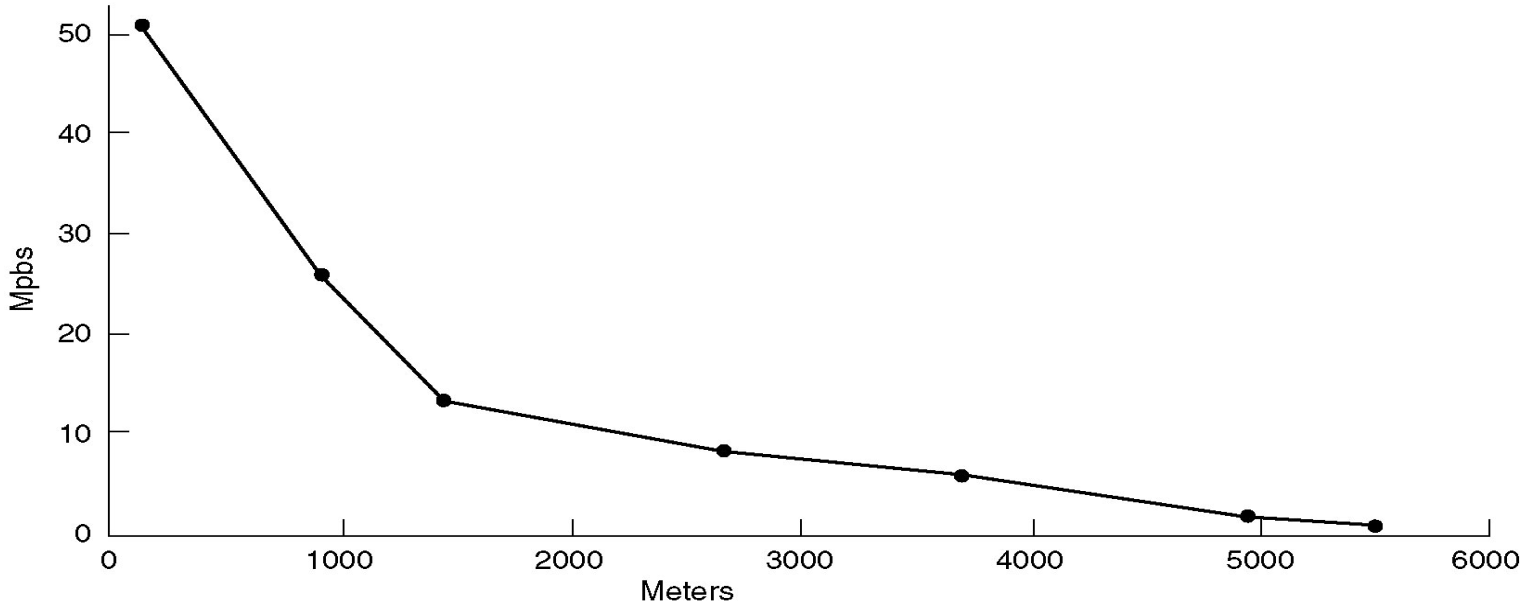
- **The data rate can be improved by removing the filter!**
  - Larger bandwidth ➔ higher data rate!

$$H \log_2 (1+S/N) \text{ bps}$$

***H*: Bandwidth**

PSTN: DSL

- **After the filter is removed, the bandwidth is determined by the physical properties of the channel**
  - Wire length, thickness, quality, etc.
  - Capacity v.s. distance for category 3 twisted pair



Longer distance ➔ larger service radius ➔ more customers ➔ more money! ☺

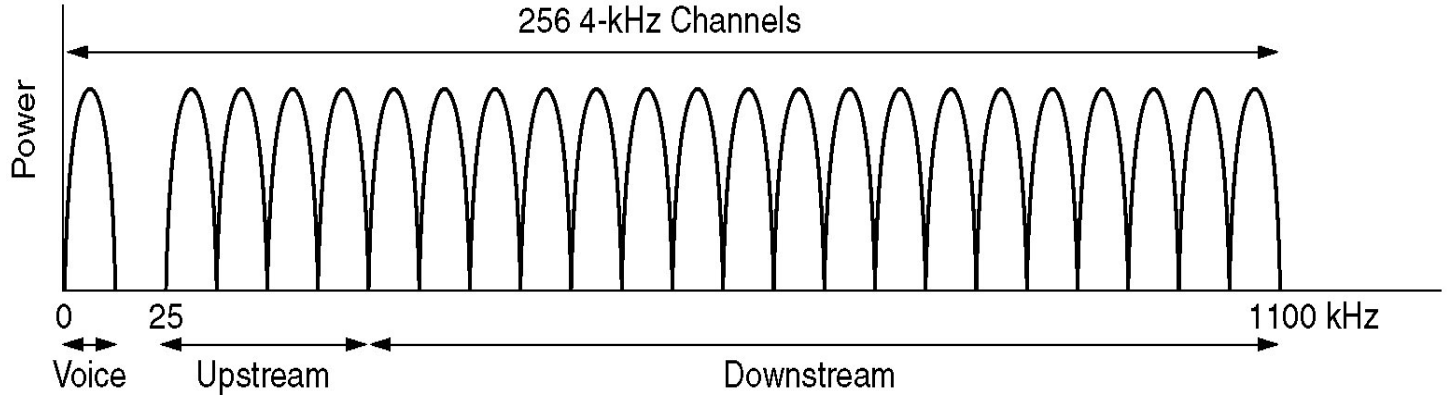
Longer distance ➔ smaller capacity ☹

PSTN: DSL

- **DSL Family**
  - ADSL (Asymmetric DSL)
    - Over a single copper pair
    - Up to 7Mbps downstream
    - 160kbps upstream
  - HDSL (High data rate DSL)
    - Over two copper pairs
    - Up to 1.5 Mbps symmetric
  - SDSL (Symmetric DSL)
    - Single loop version of HDSL
  - VDSL (Very-high-speed DSL)
    - Over shorter distance (1000 – 3000 feet)

PSTN: DSL

- **DMT (Discrete MultiTone)**
  - 1.1 MHz spectrum is divided into 256 channels with BW 4312.5 Hz each.
    - Channel 0 used for POTS (Plain Old Telephone Service)
    - Channel 1-5 not used
    - Channel 6 – 256 are used for DSL
  - Usually, 80% channels are for downstream, 20% upstream.
  - Each channel
    - 4000 baud
    - Up to 15 bits per symbol (will be smaller for worse channel condition)



PSTN: DSL

- **Example**
  - Each channel
    - 4000 baud
    - 15 bits per symbol at SNR of 30 dB
    - 8 bits per symbol at SNR of 18 dB
  - 224 downstream channels
    - Downstream data rate

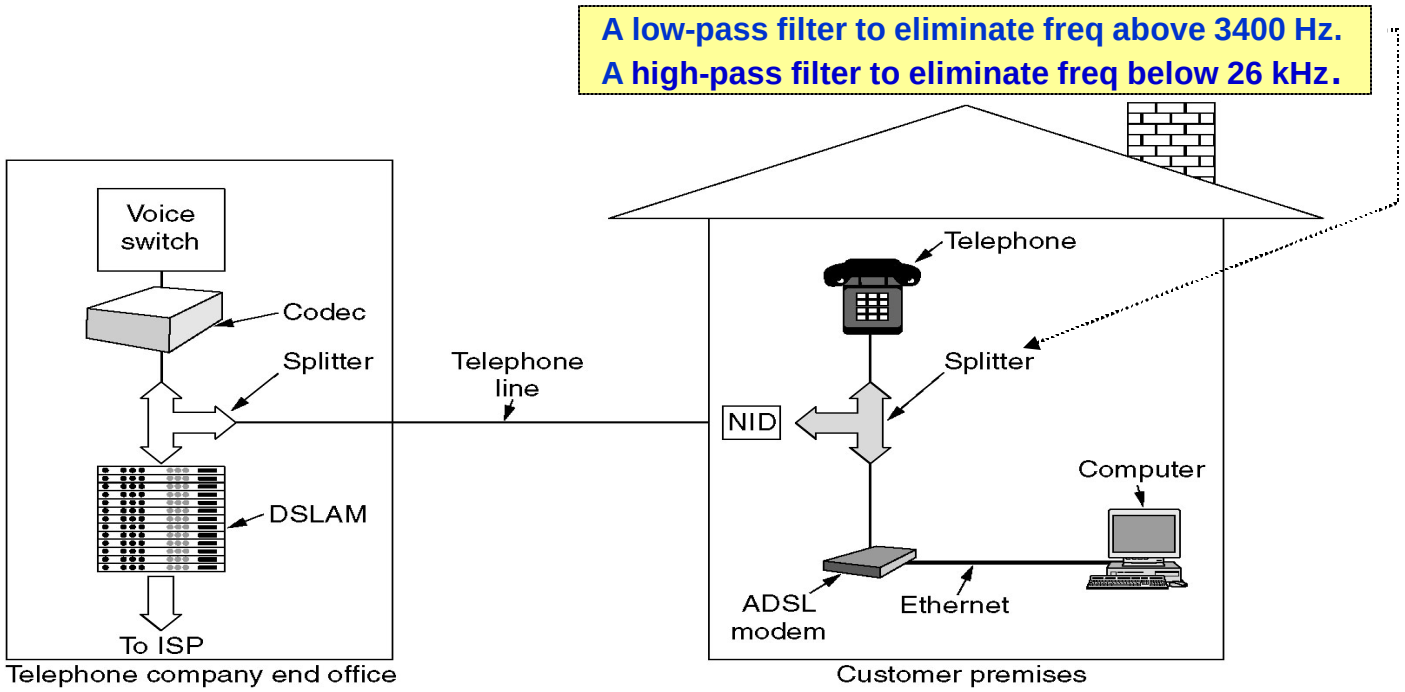
      SNR > 30 dB ➔ 4000 x 15 x 224 = 13.44 Mbps

      SNR > 18 dB and SNR < 30 dB ➔ 4000 x 8 x 224 = 7.168 Mbps

Longer distance ➔ smaller SNR ➔ smaller data rate

PSTN: DSL

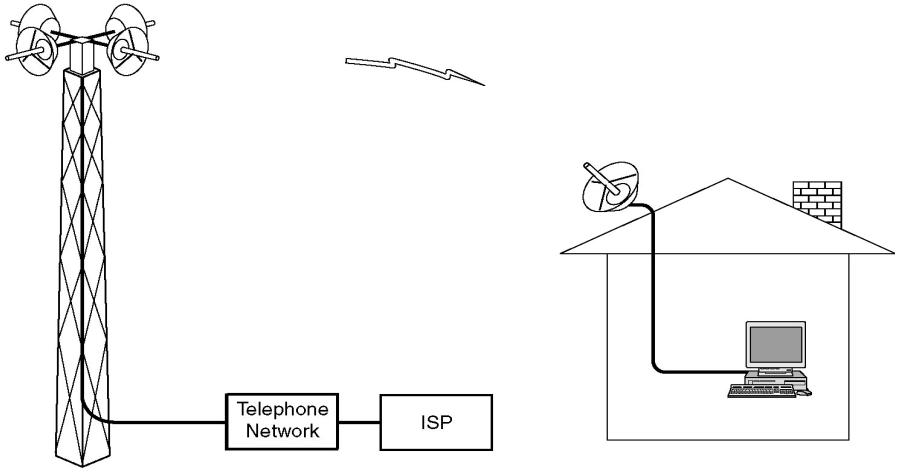
- **A typical ADSL arrangement**
  - NID: network interface device
    - Required by ITU G.992.1 (up to 8Mbps down stream)
    - Not required by ITU G.992.2 (up to 1.5Mbps down stream)
  - DSLAM: Digital subscriber line access multiplexer



PSTN: WIRELESS LOCAL LOOP (WLL)

- **Standards**
  - MMDS (Multichannel Multipoint Distribution Service)
  - LMDS (Local Multipoint Distribution Service)
    - Range: 2-5 km. DS: 36 Mbps, US: 1Mbps
    - Require line-of-sight (LOS) ➔ high tower
  - WiMax (Worldwide Interoperability for Microwave Access)
    - Range: 50 km. 70 ~ 80Mbps (not fixed yet).
    - Doesn't require LOS.

LMDS example

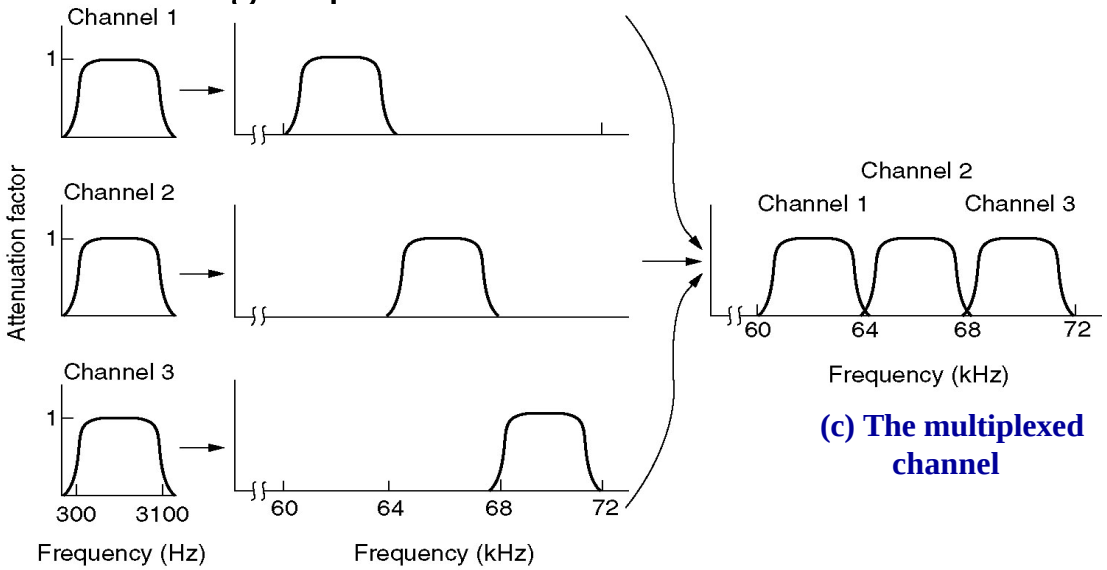


PSTN: TRUNKS AND MULTIPLEXING

- **Multiplexing**
  - Carry different services/conversations over a single physical trunk.
- **Multiplexing Schemes**
  - Frequency Division Multiplexing (FDM)
    - Frequency spectrum is divided into multiple frequency bands.
    - Each user/service is assigned one or more bands.
  - Time Division Multiplexing (TDM)
    - Time is divided into slots
    - Each user is assigned one or more slots.

PSTN: MULTIPLEXING

- **Frequency Division Multiplexing**
  - Telephone channel: 3100 Hz (300 Hz ~ 3400 Hz)
  - Bell system trunks
    - Group: 12 voice channels
    - Super group: 60 voice channels
    - Master group: 300 voice channels

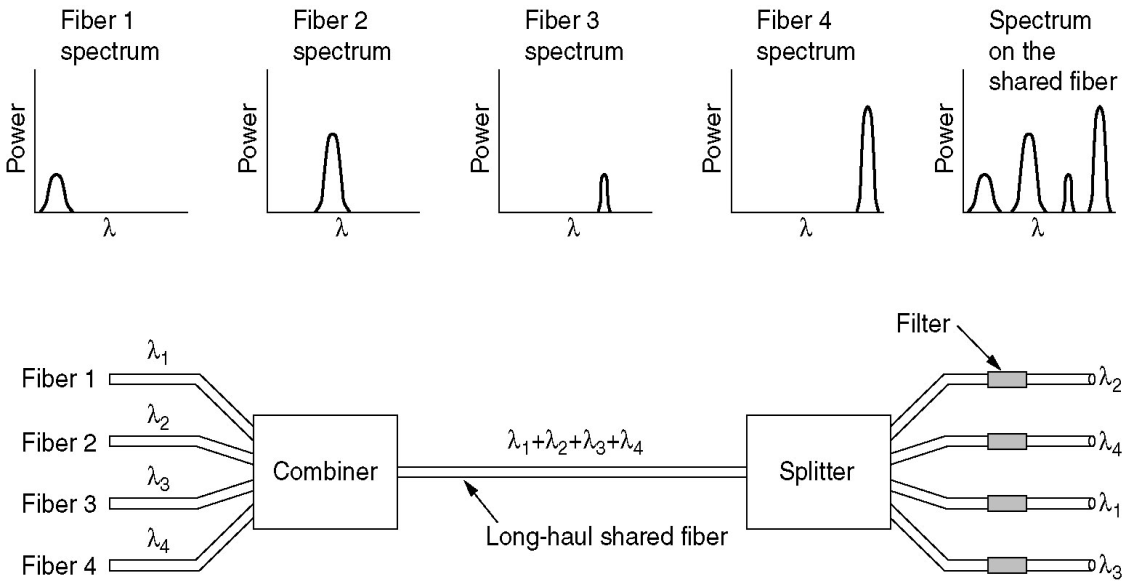


(a) The original bandwidths **(b) Bandwidths raised in frequency**

(c) The multiplexed channel

PSTN: MULTIPLEXING

- **Wavelength Division Multiplexing (WDM)**
  - Given speed of waveform, there is a one-to-one relationship between frequency and wavelength $l = c / f$
  - WDM is just an variation of FDM on fiber at very high frequencies!
    - Operates in 1550 nm range, 50 GHz spacing
    - ITU-T G.692 specifies 50-GHz x 80 CHs



PSTN: MULTIPLEXING

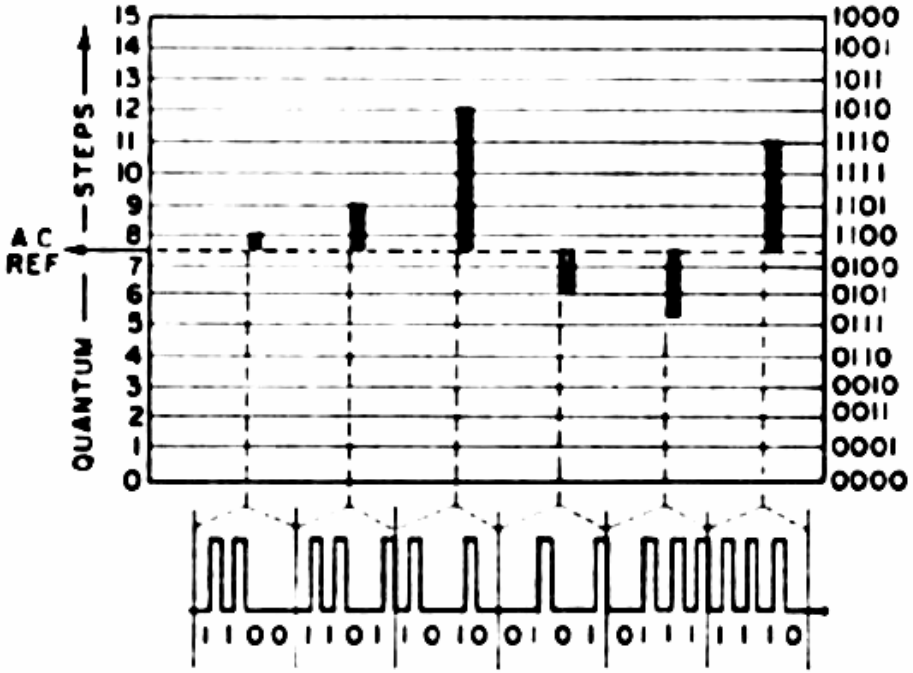
- **Time Division Multiplexing (TDM)**
  - Time is divided into slots, with each user assigned one or more slots.
  - Only suitable for digital system.
    - Digital data can be easily cut into pieces.
- **TDM is used in PSTN backbone.**
  - Most local loops carry analog voice.
- **How to convert analog voice from local loop to digital data at end office?**
  - Pulse code modulation (PCM)
  - Differential pulse code modulation (DPCM)
  - Delta Modulation (DM)
  - …

PSTN: MULTIPLEXING

- **Pulse Code Modulation**
  - Sampling: 8000 Samples/second (125 us/sample)
  - Quantization: represent each sample by 8-bit sequence (256 discrete levels)
  - Data rate: 8000 x 8 = 64 kbps



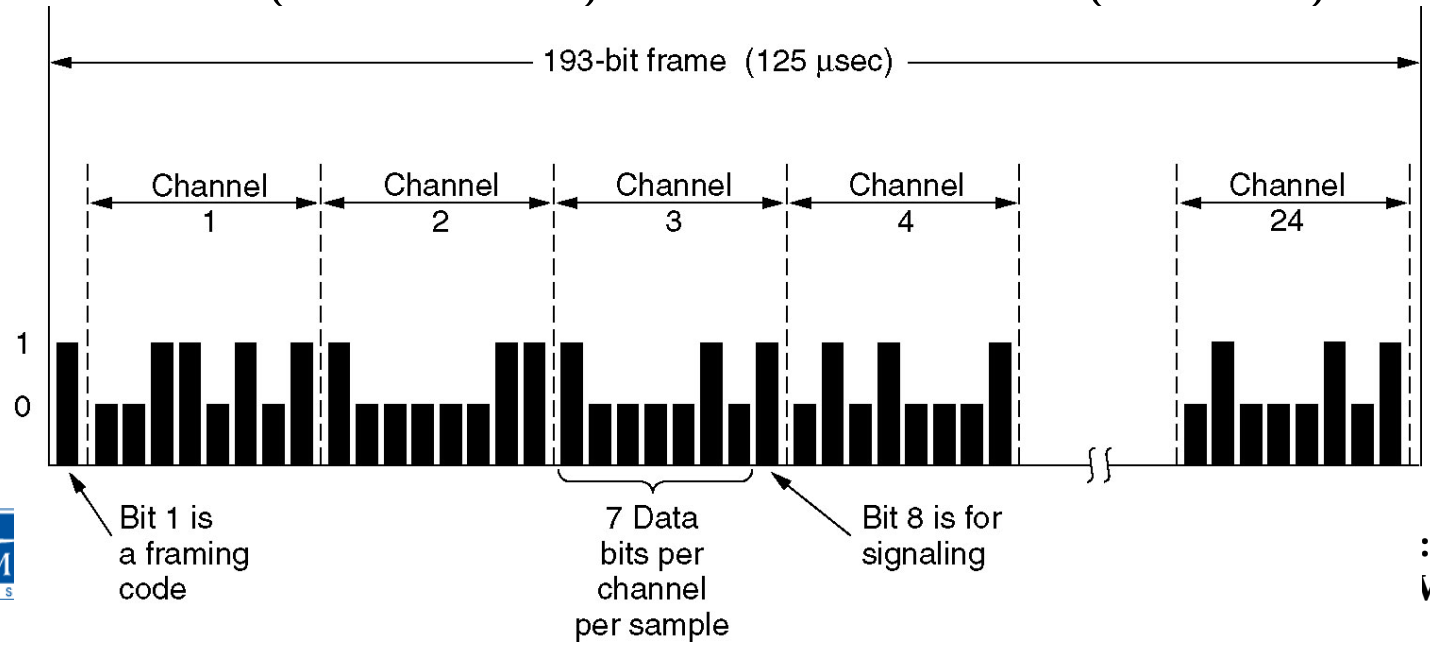
Sampling



Quantization

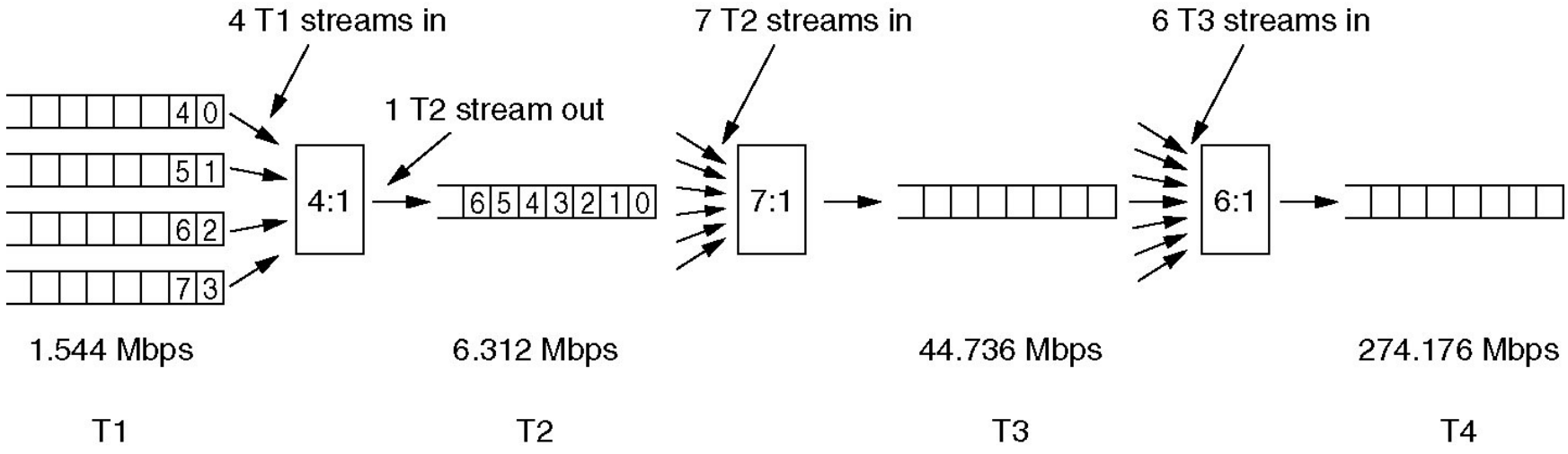
PSTN: MULTIPLEXING

- **Example: T1 carrier**
  - 24 voice channels (slots) multiplexed together
  - Each channel (slot) has 8 bits
    - 7 bits information, 1 bit signaling
  - The total duration of the 24 channels is 125 us
    - The channel for one user will appear 1/125us=8000 times per second
  - Data rate for one channel: 8000 x 7 = 56 kbps.
  - Gross rate (data + control) for T1 carrier: 8000 x (8 x 24 +1)=1.544Mbps.



PSTN: MULTIPLEXING

- **Multiplexing T1 into higher order carriers**
  - Multiplexer arrange the streams from lower rate carrier in a round-robin manner
  - Europe use different scheme: E1, E2, E3, …

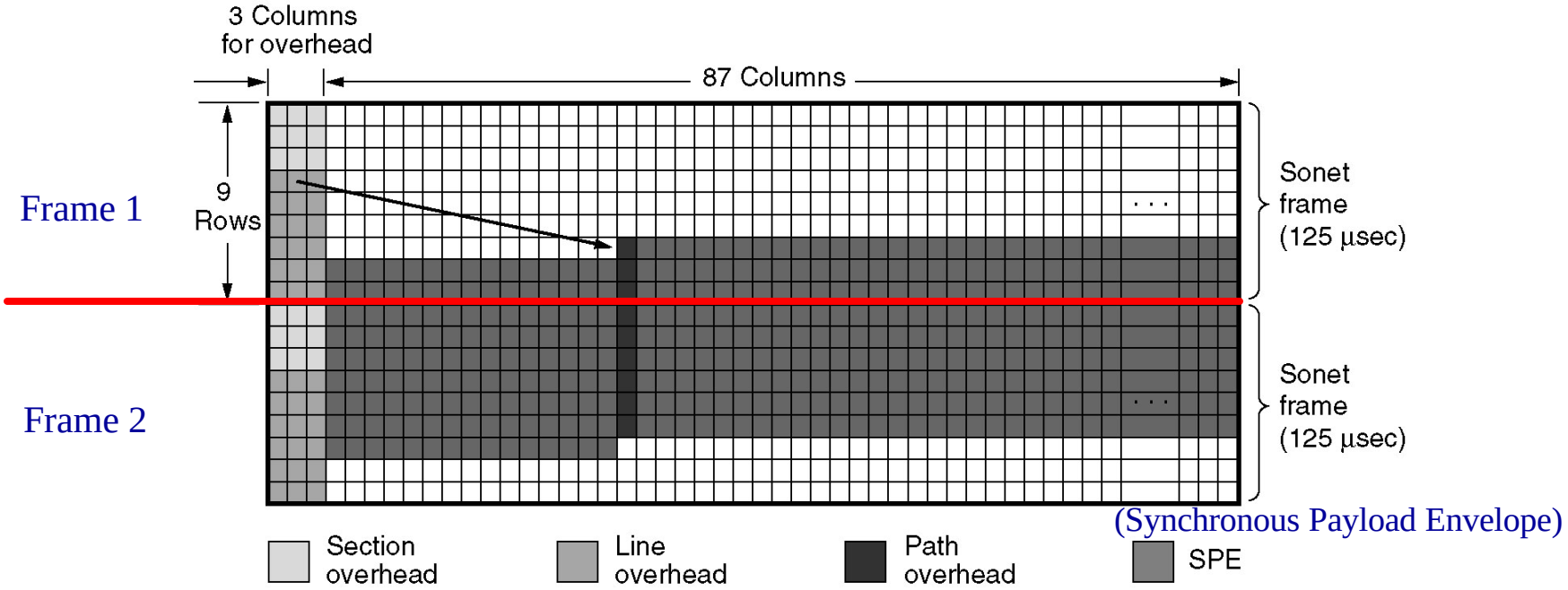


PSTN: MULTIPLEXING

- **SONET/SDH**
  - SONET (Synchronous Optical Network): US standard.
  - SDH (Synchronous Digital Hierarchy): International standard.
  - Describes the TDM format on fiber optics.
- **Synchronous!**
  - Sender and receiver are tied to a common clock.
  - Bits are grouped into frames
  - Frames are sent out back-to-back ➔ no gap between frames
  - Each frame is sent out at extremely precise interval
    - Unlike ATM (Asynchronous Transfer Mode), where cell can be sent out at any time.
  - When no data to Tx, send out empty (dummy) frame!

PSTN: MULTIPLEXING

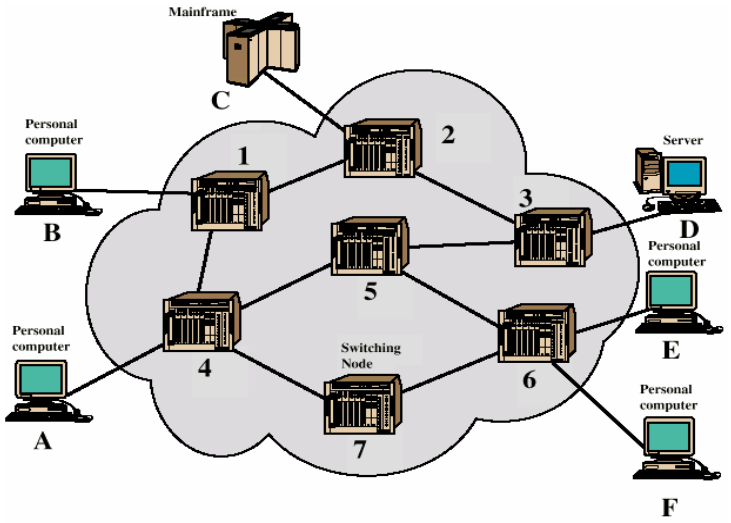
- **Basic SONET frame structure**



  - Frame duration: 125 us. Each frame contains 9 x 90 = 810 bytes, where 87 x 9 = 783 bytes of them are for user data (payload)
    - Gross rate: 810 bytes / 125 us x 8 bits/byte = 51.84 Mbps
    - Net data rate: 783 bytes / 125 us x 8 bits/byte = 50.112 Mbps

PSTN: SWITCHING

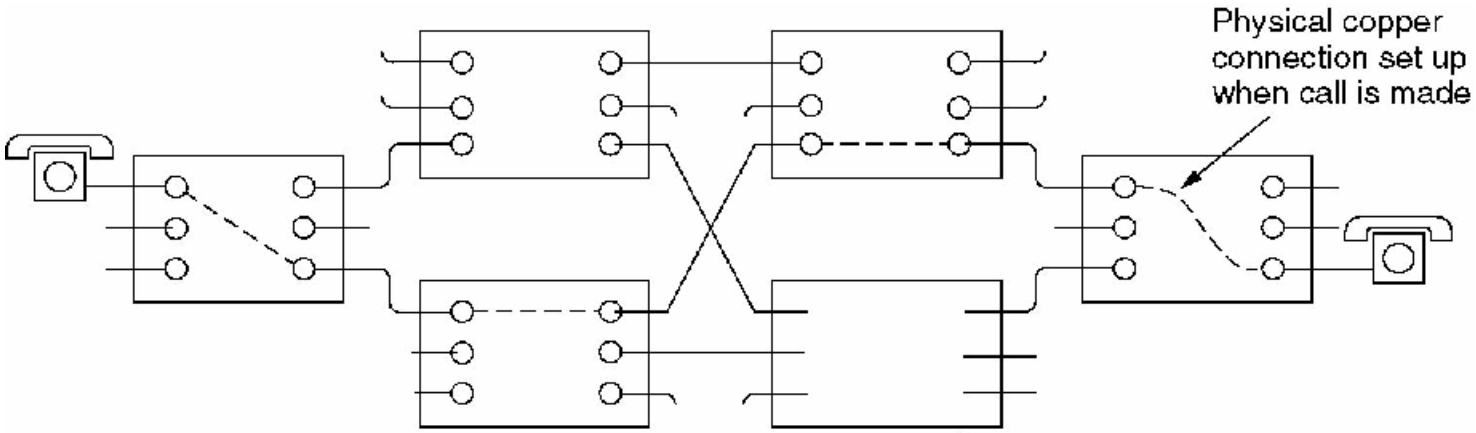
- **Long distance transmission between two hosts usually relies on subnet**
  - Subnet is a collection of Tx lines and switching elements
  - Data are routed by being switched from one switching element to another



- **Three switching techniques**
  - Circuit Switching
  - Message Switching
  - Packet Switching

PSTN: SWITCHING

- **Circuit Switching**
  - The switching elements in subnets seek and reserve a physical path between the communication hosts.
  - The physical path is completely dedicated to the communication.
  - An end-to-end path needed to be set up before data Tx
    - Set up time: time elapsed after end of dialing and before ringing
      - Could up to several seconds for long distance call



PSTN: SWITCHING

- **Circuit Switching Characteristics**
  - Three phases
    - Circuit set up, data transfer, circuit release (disconnect)
  - Must have switching capacity to establish connection
  - Once the circuit is set up, the only delay is the propagation time of the electro-magnetic signal
    - No store and forward
    - No error correction
  - All the data are transferred follow exact the same path.
  - Overhead at connection set up, no overhead during transfer
  - Most suitable for voice communication (or continuous data)
    - Minimum delay during call
    - Resource completely dedicated to a particular call
    - Not efficient for burst data

PSTN: SWITCHING

- **Message Switching**
  - The entire message is received and stored by a switching element.
  - Error corrections are performed at switching element
  - The entire message is forwarded to the next switching element.
- **Shortcomings**
  - One lengthy message can take up the Tx line for a very long time.
    - Long queuing time
    - Not good for delay sensitive or interactive services
  - The size of message is usually large
    - Usually disk or tape are required at the switching elements to store message
      - Long delay.

PSTN: SWITCHING

- **Packet Switching**
  - Message are divided into shorter packets
    - Strict upper limit on packet size
  - Each packet is received by each switching element in its entirety.
  - Packet is stored at switching element
  - Packet is forwarded to the next switching element
- **Advantages over message switching**
  - A packet will not take up a line for too long
  - Packet can be stored in buffer instead of disks or tapes.
  - The first packet can be sent out before the second packet arrives
    - Reducing delay
    - Improving throughput

PSTN: SWITCHING

- **Circuit Switching v.s. Packet Switching**

Item	Circuit-switched	Packet-switched
Call setup	Required	Not needed
Dedicated physical path	Yes	No
Each packet follows the same route	Yes	No
Packets arrive in order	Yes	No
Is a switch crash fatal	Yes	No
Bandwidth available	Fixed	Dynamic
When can congestion occur	At setup time	On every packet
Potentially wasted bandwidth	Yes	No
Store-and-forward transmission	No	Yes
Transparency	Yes	No
Charging	Per minute	Per packet

In addition, circuit switching is good for QoS, packet switching is more fault tolerant.

OUTLINE

- Public Switched Telephone Network (PSTN)
- **Mobile Telephone System**
- Cable Television

MOBILE PHONE

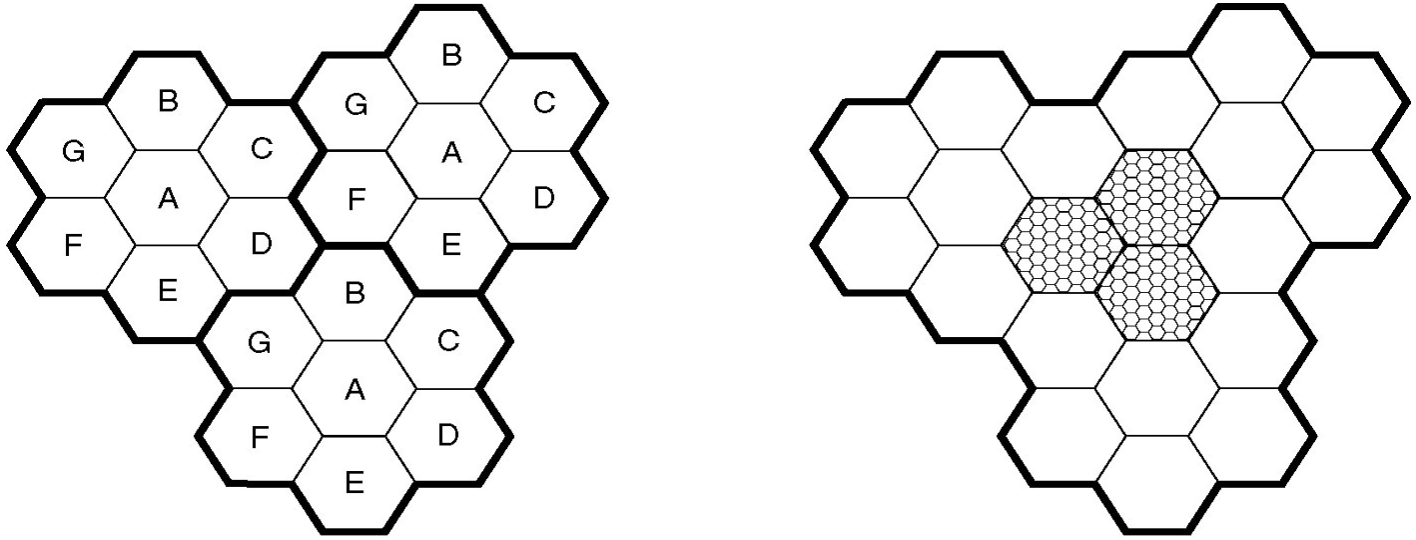
- **Mobile phone has gone through three generations**
  - First generation: Analog voice
    - Example: AMPS
  - Second generation: Digitalized voice and narrowband data
    - Examples: GSM, D-AMPS, IS-95 (Interim Standard, CDMA)
  - Third generation: Digitalized voice and broadband data
    - Examples: UMTS, cdma2000

MOBILE PHONE: 1ST GENERATION

- **AMPS**
  - Advanced Mobile Phone System
  - Launched in U.S. in 1982 by AT&T
  - The 1st generation cell phone system
  - Analog voice
    - but the control information (such as call set up) are digital.
- **Frequency Division Multiple Access (FDMA)**
  - Each user is dynamically assigned a different frequency for communication
  - There are totally 832 full-duplex channels
    - BW/Channel: 30KHz downlink, 30KHz uplink
  - How can we use 832 channels to support the simultaneous communication of practically unlimited number of users?
    - Cell Concept with Frequency Reuse

MOBILE PHONE: 1ST GENERATION

- **Cell Concept with Frequency Reuse**
  - The entire coverage area is divided into cells
  - A base station (BS) in the center of each cell.
  - Communication occurs between BS and Mobile Stations (MS) in cell.
  - The same frequency (channels) can be reused by nearby cells ➔ Frequency Reuse
    - But not by adjacent cells!➔Reduce co-channel interference (CCI).



(a) (b)

MOBILE PHONE: 1ST GENERATION

- **Advantages of Frequency Reuse**
  - Limited frequency resource can be reused to support unlimited number of users
    - Each cell support only limited number of users
    - The number of users supported by system increase linearly with the number of cells
  - Each BS only covers one cell
    - Smaller Tx power at BS ➔ Less interference to other systems
    - Smaller Tx power at MS ➔ Longer battery life
  - A cell can be further divided into smaller cells to support more users
    - Upgradability
- **Cell Radius in AMPS: 10 ~ 20 km**

MOBILE PHONE: 1ST GENERATION

- **Channels**
  - Uplink: transmission from MS to BS
    - 832 simplex channels for uplink in 824 MHz ~ 849 MHz
    - BW of each simplex channel: 30 kHz
  - Downlink: transmission from BS to MS
    - 832 simplex channels for downlink in 869 MHz ~ 894 MHz
    - BW of each simplex channel: 30kHz
  - Downlink channel + uplink channel = full-duplex channel
    - There are totally 832 full-duplex channels
- **Channel Allocation**
  - Data channel (bi-directional): voice, fax, or data
  - Control channel (BS to MS): system control
  - Access channel (bi-directional): call setup
  - Paging channel (BS to MS): alert MS to calls for them
- **Typically, 45 data channels in each cell.**

MOBILE PHONE: 1ST GENERATION

- **MSC (Mobile Switching Center)**
  - One MSC is connected to multiple BSs
  - Serving as end office of PSTN
  - Each MSC is connected to at least one switching office of PSTN.
    - Cell phone users can talk with residential phone users.
- **Call Management**
  - Each MS needs to register with a BS based on its location.
    - When a phone is turned on, it scans for control channels
    - Broadcast its own identity (phone # and serial #)
    - BS will forward the identity information to MSC
    - MSC registers the existence of the MS, informs the MS's home MSC of its current location.
    - MS registers once every 15 minutes.

MOBILE PHONE: 1ST GENERATION

- **Call Management (Cont'd)**
  - Making a call
    - MS send its own identity and recipient # to BS
    - BS forwards call request to MSC
    - MSC finds an idle data channel.
      - If no channel available, the call will not get through.
    - BS informs MS the channel #, and MS switch to that channel.
  - Incoming call
    - MS continuously listen to paging channel
    - When a call is placed to a MS, a message is first sent to its home MSC.
    - Home MSC will forward the information to its currently serving MSC and BS.
    - Serving BS broadcast the call request to all MS through paging channel
    - MS answers the request
    - BS allocate channel for the call.

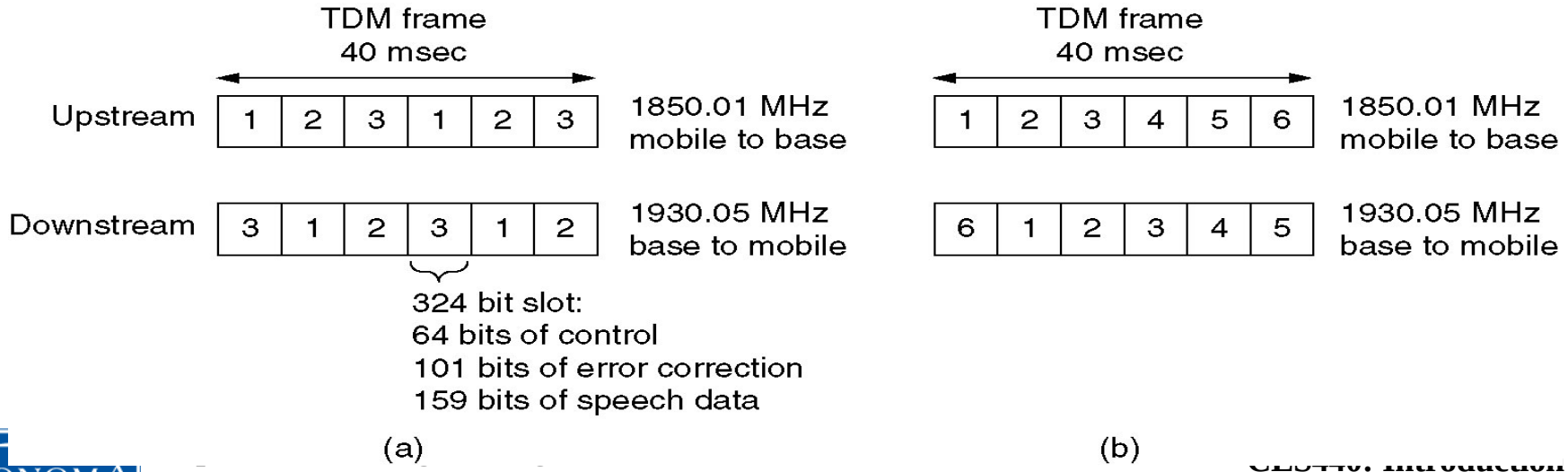
MOBILE PHONE: 2ND GENERATION

- **Major differences from 1st generation system: digitalized voice**
  - Pure digital communication
  - Better performance
- **Many different standards**
  - D-AMPS (Digital AMPS, two versions IS-54, IS-136)
    - Cingular, AT&T
  - GSM (Global System for Mobile Communication)
    - Europe, T-Mobile, Cingular, AT&T
  - IS-95 (using CDMA technology, also called CDMAONE)
    - Sprint
  - PDC (Personal Digital Cellular)
    - Japan (NTT DoCoMo)

IS: Interim Standard

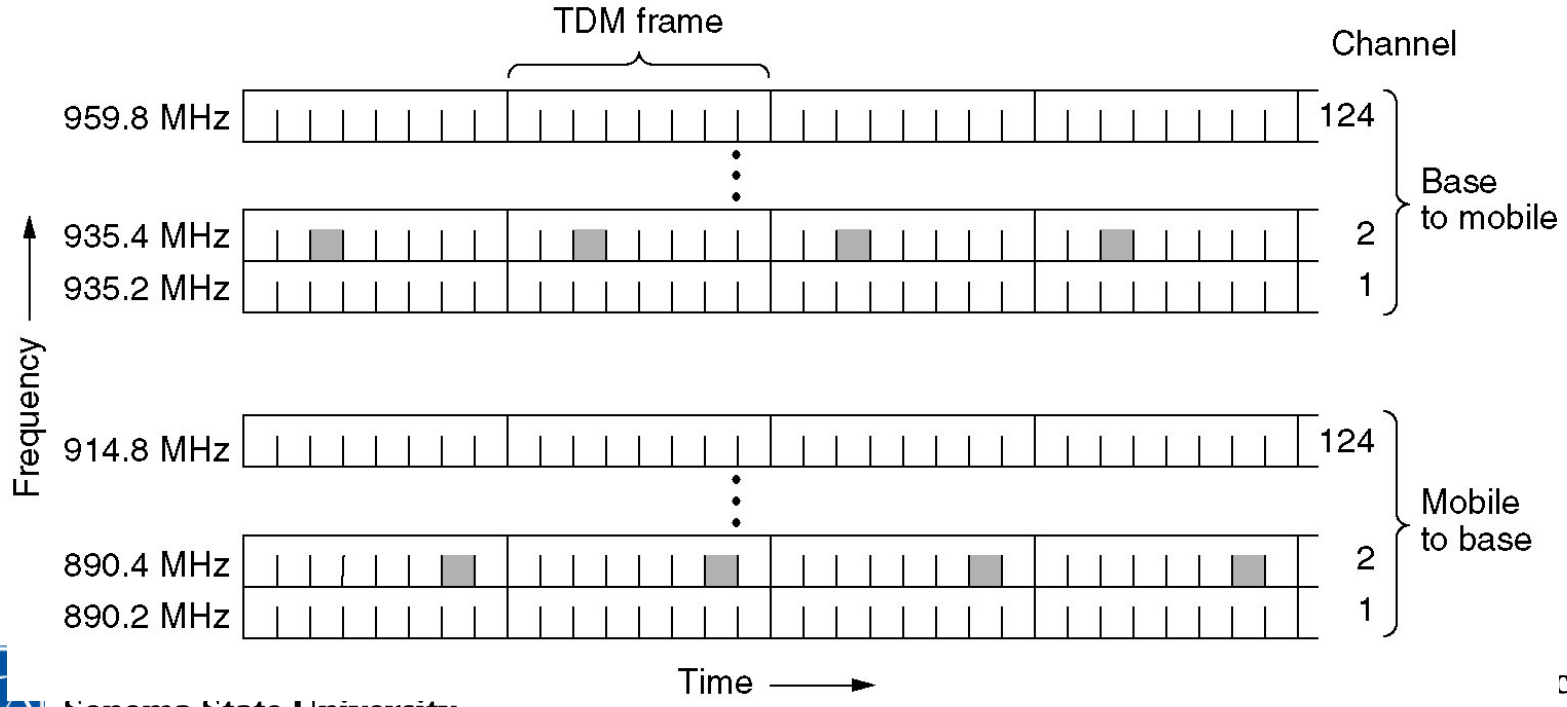
MOBILE PHONE: 2ND GENERATION

- **D-AMPS**
  - Analog voice is digitalized and compressed by vocoder before Tx.
  - 30 kHz channel
    - Can coexist with AMPS
- **TDMA is used in each channel**
  - Time Division Multiple Access
  - The time is divided into slots, users take turns to use different slots
  - Multiple users share the same channel ➔ capacity multiplied!



MOBILE PHONE: 2ND GENERATION

- **GSM (Global System for Mobile communication)**
  - The most widely used 2nd generation system
  - Similar to D-AMPS, but with different format
  - 124 full-duplex frequency channels
  - BW of each frequency channel: 200kHz
  - TDMA is used in each channel: 8 users can be supported



MOBILE PHONE: 2ND GENERATION

- **IS-95 (CDMA)**
  - Code Division Multiple Access
  - All users Tx simultaneously using the same frequency.
  - Different codes are used to separate the signals from different users.
- **Each user is assigned a unique N-bit code: chip sequence**
  - E.g., the code for one user is c = [-1 -1 -1 1 1 -1 1 1]
  - To transmit '1', it will send out [-1 -1 -1 1 1 -1 1 1]
  - To transmit '0', it will send out [1 1 1 –1 –1 1 –1 –1]
  - At receiver, performing inner product between rx sequence and code of the user

$$\mathbf{a} = [a_1, a_2, \cdots, a_N] \qquad \mathbf{b} = [b_1, b_2, \cdots, b_N]$$

$$\mathbf{a} \bullet \mathbf{b} = \frac{1}{N}(a_1 b_1 + a_2 b_2 + \cdots + a_N b_N)$$

  - For binary vector **a** (contains either 1 or –1), $\mathbf{a} \bullet \mathbf{a} = \frac{1}{N}\sum_{k=1}^{N}(\pm 1)^2 = 1$

MOBILE PHONE: 2ND GENERATION

- **Codes belonging to different users are orthogonal to each other**
  - If $\mathbf{a} \bullet \mathbf{b} = 0$, then **a** and **b** are orthogonal to each other.
- **CDMA example**
  - Code for user 1: $\mathbf{c}_1 = [-1,-1,-1,+1,+1,-1,+1,+1]$
  - Code for user 2: $\mathbf{c}_2 = [-1,-1,+1,-1,+1,+1,+1,-1]$
  - User 1 Tx '1', signal sent out by user 1 is $\mathbf{s}_1 = [-1,-1,-1,+1,+1,-1,+1,+1]$
  - User 2 Tx '0', signal sent out by user 2 is $\mathbf{s}_2 = [+1,+1,-1,+1,-1,-1,-1,+1]$
  - The received signal is $\mathbf{r} = \mathbf{s}_1 + \mathbf{s}_2 = \mathbf{c}_1 - \mathbf{c}_2 = [0,0,-2,+2,0,-2,0,+2]$
  - To detect signal from user 1 $\mathbf{c}_1 \bullet \mathbf{r} = (0+0+2+2+0+2+0+2)/8 = 1$
  - To detect signal from user 2 $\mathbf{c}_2 \bullet \mathbf{r} = (0+0-2-2+0-2+0-2)/8 = -1$
  - Why?

$$\mathbf{c}_1 \bullet \mathbf{r} = \mathbf{c}_1 \bullet (\mathbf{c}_1 - \mathbf{c}_2) = \mathbf{c}_1 \bullet \mathbf{c}_1 - \mathbf{c}_1 \bullet \mathbf{c}_2 = \mathbf{c}_1 \bullet \mathbf{c}_1 = 1$$

$$\mathbf{c}_2 \bullet \mathbf{r} = \mathbf{c}_2 \bullet (\mathbf{c}_1 - \mathbf{c}_2) = \mathbf{c}_2 \bullet \mathbf{c}_1 - \mathbf{c}_2 \bullet \mathbf{c}_2 = -\mathbf{c}_2 \bullet \mathbf{c}_2 = -1$$

MOBILE PHONE: 2ND GENERATION

- **CDMA is also called Spread Spectrum**
  - Chip period << symbol period ➔ Radio signals change faster in a unit time ➔ Much larger bandwidth compared to original signal
- **Advantages of CDMA**
  - Better noise and interference immunity
    - Less SNR requirement
  - Larger capacity
  - Better security
- **Disadvantages of CDMA**
  - Larger bandwidth
  - Interference from other users
    - Users are not synchronized to each other

MOBILE PHONE: 2.5 GENERATION

- **2.5G System**
  - EDGE: Enhanced Data Rates for GSM Evolution
    - Similar to GSM, but using larger modulation alphabet size ➔ more bits carried by one symbol
  - GPRS: General Packet Radio Service
    - A packet switching network on top of D-AMPS or GSM
- **Both of the two systems are just enhancement of 2G systems**
  - More suitable to be called 2.1G.

MOBILE PHONE: 3RD GENERATION

- **Objectives**
  - Integrated voice and broadband data
  - One standard serves the whole world
    - This objective is not achieved
    - today 2 main 3G standards co-exist
- **The general name for 3G: IMT-2000**
  - International Mobile Telecommunication 2000
  - Project launched by ITU in 1992
  - Different countries have different proposals
  - So far, two survivors
    - W-CDMA (Wideband CDMA), also called UMTS (Universal Mobile Telecommunication System), proposed by Ericsson.
    - cdma2000, proposed by Qualcomm.

MOBILE PHONE: 3RD GENERATION

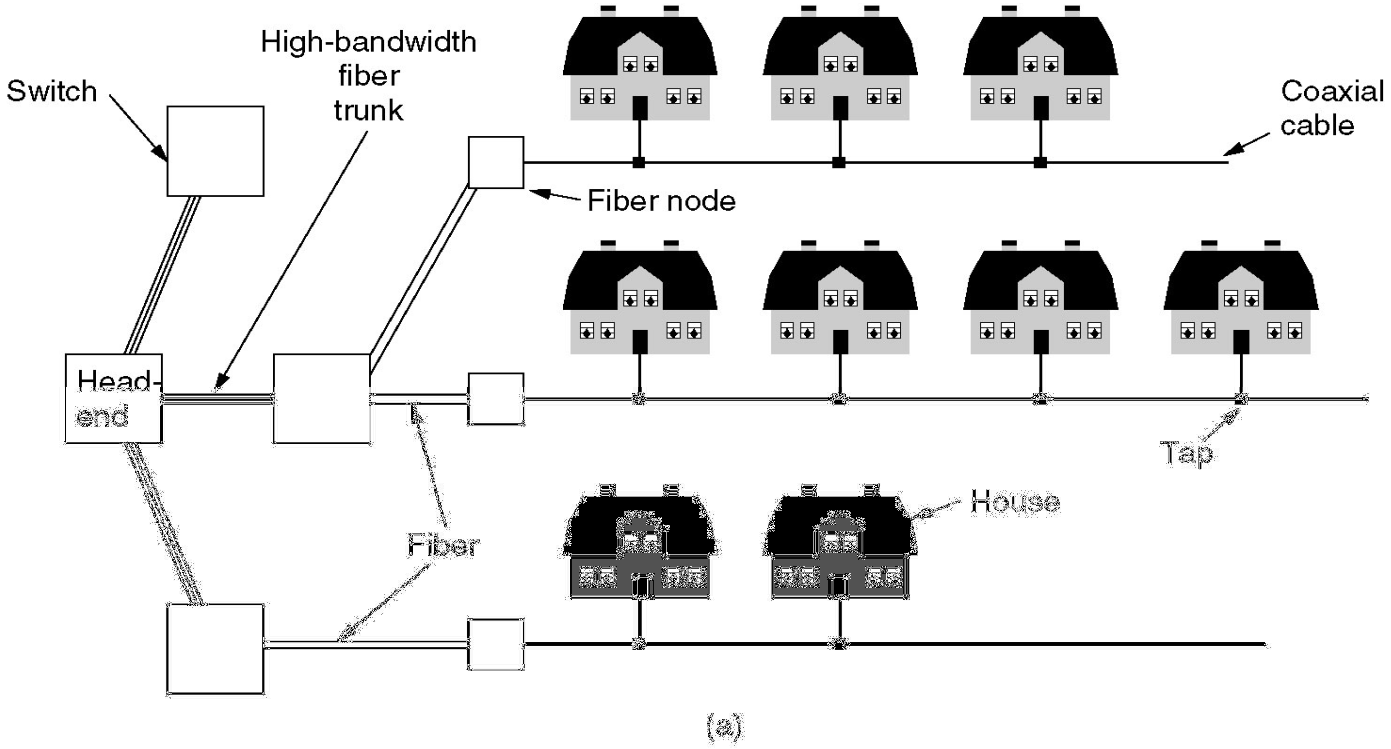
- **UMTS and cdma2000 are very similar to each other**
  - CDMA
  - Channel BW: 5MHz
  - Only some detailed technical differences
- **After 1998, the standard activities of 3G system are organized by 3GPP**
  - Third Generation Partnership Project
  - UMTS: www.3gpp.org
  - cdma2000: www.3gpp2.org
- **Today**
  - Standards are still under development (was planned to launch at 2000)
  - Some service providers start to provide partial 3G services
    - Verizon, Sprint, etc.

OUTLINE

- Public Switched Telephone Network (PSTN)
- Mobile Telephone System
- **Cable Television**

CABLE TELEVISION: STRUCTURE

- **HFC (Hybrid Fiber Coax) System**
  - Fiber used for long-haul runs
  - Coaxial cable to the house
  - Initially designed for one-way service.

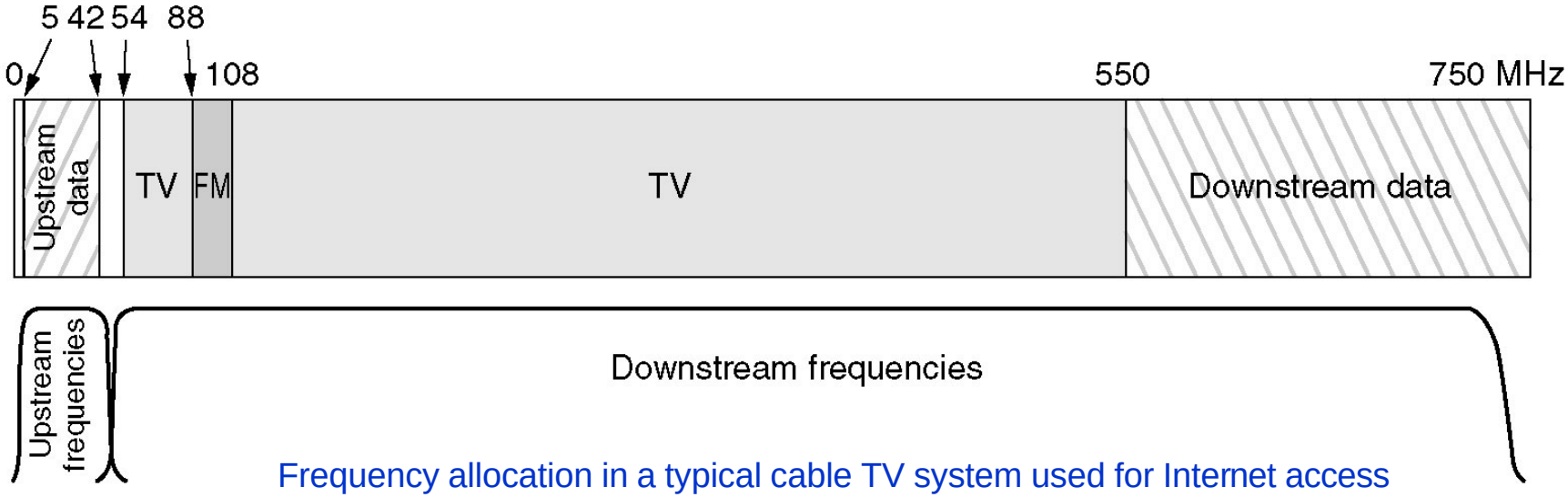


CABLE: INTERNET OVER CABLE

- **Problems & Solutions**
  - Most amplifiers in system are one-way
    - Only amplify signals from head end to user
    - Solution: replace them by two way amplifiers
  - Many users share the same coaxial cable
    - In telephone system, each house has its own local loop
    - In HFC system, all users connecting to the same cable must share the limited bandwidth resource
    - Solution: split long cables to shorter ones and connecting them directly to fiber node.
      - Reduce the # of users sharing the same cable

CABLE

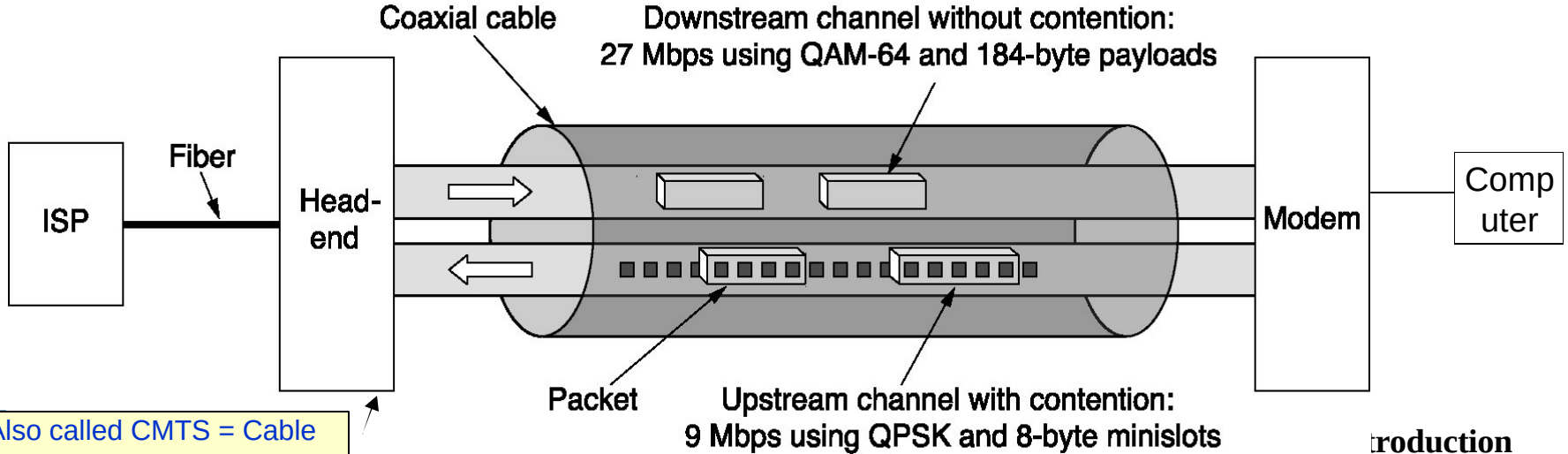
- **Spectrum Allocation**
  - Upstream (user to headend): 5 ~ 42 MHz
    - Modulation: 2 bits/symbol (QPSK)
  - Downstream (headend to user): 550 ~ 750 MHz
    - Modulation: 6 bits/symbol (64QAM), or 8 bits/symbol (256QAM)



Frequency allocation in a typical cable TV system used for Internet access

CABLE

- **Standard: DOCSIS**
  - Data Over Cable Service Interface Specification
- **Upstream**
  - All users sharing one cable contending for the resource (TDMA)
  - Collisions could occur – ALOHA is employed
- **Downstream**
  - Only one transmitter: Headend
  - Headend controls the transmission in downstream➔no contention



Also called CMTS = Cable Modem Termination System

CABLE V.S. DSL

	ADSL	Cable
Physical Medium at backbone	Fiber	Fiber
Physical Medium at access	Twisted pair	Coax cable
Carrying capacity	~1 Mbps DS, ~256 kbps US	Depends upon number of users, Theoretically many time ADSL
Performance	Predictable	Unpredictable, depends on # of users
Security	Secure (private connection)	Not secure (shared access), needs encryption
Availability	Depends upon ADSL service availability	Depends upon cable service availability
Reliability	More reliable, PSTN has backup power	Not as reliable, power amplifiers may loose power
Choice of ISP	ADSL provider may provide choice of ISP	Cable provider hardly provides choice of ISP

SUMMARY

- **Telephone Systems**
  - Local loops, trunks, switches
  - Dial-up, ADSL, WLL
- **Mobile Phone Systems**
  - 1G (AMPS)
    - Analog voice
    - FDMA
  - 2G
    - Digitalized voice, narrow band data
    - TDMA (GSM, D-AMPS), CDMA (IS-95)
  - 3G
    - Digitalized voice, broadband data
    - CDMA
- **Internet over cable: HFC**
  - Users sharing cable ➔ users competing for the resource.